Memo: BASF process column K 2750 (DEXPI C02, unseen)

2026-11-02 16:01 +03 · DRAFT, not approved

Decision asked for: +20 % feed: what limits it first and what does it cost?

Recommendation: Simulator screening passed; proceed to data and vendor verification (not ready for approval): 1 change to reach 120 % of today's throughput - (1) Replace column with 600 mm ID rated shell (vendor to confirm).

Result: At 120 %, existing plant: 1 limit over 100 % (highest K 2750 distillation Fair flood fraction vs limit 110 %); with the changes: all modelled process limits within 100 % (highest K 2750 distillation Fair flood fraction vs limit 76 %). Simulated maximum throughput: 113.1 % → 145.1 % of today.

- (1) Replace column with 600 mm ID rated shell (vendor to confirm) - cost class not stated

Risk: no second-opinion disagreement recorded. Relief findings on changed equipment: 1.

Measure next: Actual product purity spec for 2750.001.6 and *at N3* vent/distillate; Feed rate/composition envelope and actual reboiler/condenser capacity; Tray type, spacing and fouling/pressure-drop.

Full MOC file with Rev A/Rev B drawings and PSSR checklist (6 items) attached. Audit head `3991f13052d3`.